AMAN LONARE

Email: amanlonare95@gmail.com | Phone: +81-80-7557-2462 | Location: Tokyo, Japan | Github | Portfolio

SUMMARY

AI and Machine Learning Engineer with 5+ years of experience building & deploying production-grade ML and LLM systems. Strong background in designing end-to-end AI platforms and automated AI workflows, including data pipelines, model training, scalable inference, and MLOps, delivering reliable, high-performance production solutions.

TECHNICAL SKILLS

Programming & Development: Python, Go, SQL, Bash, Dart, React, FastAPI, Flask, Cursor, Claude Code
AI and ML: TensorFlow, PyTorch, Hugging Face, ETL Pipeline, A/B Testing, RLHF, DSPy, PydanticAI
LLMs: LangGraph, RAG, Deep Agents, OpenAI, MCP, Vector/Graph Database, DPO/PEFT, Quantization
MLOps: AWS SageMaker & Bedrock, MLflow, Airflow, CI/CD, Model Monitoring, LLMOps, Celery, LangSmith
Cloud & Tools: Docker, Kubernetes, Terraform, AWS EMR, Git, Vercel, Apache Spark, Flutter
Communication: English (Fluent), Hindi (Fluent), Japanese (N5, *N4 candidate (Expected [06/2026])*)

PROFESSIONAL EXPERIENCE (5+ YEARS)

AI Engineer | Archetype Studio, JPN | Full Time *Aug 23 – Present*

- **Project: Multi-Agent Orchestration for End-to-End Engineering *(in progress)***
  - Implementing **Multi-Agent Orchestration** using **LangGraph** & **Bedrock** for automated planning & code generation with a **Centralized Supervisor** to route tasks between specialized agents (Planning, Coder, Ops)
  - Integrating **Human-in-the-Loop** gates to ensure high-fidelity output and safety during automated cycles
- **Project: AI Coding Assistant for a FinTech Platform (Client: Omise Japan)**
  - Architected and implemented core developer pipelines enabling end-to-end AI workflows, including **Figma-to-Code**, **Advanced RAG** ingestion, **API and schema** tooling, **execution runners**, and **plugin SDKs**
  - Built agentic architectures with **deep agents**, & specialized coordinators to orchestrate coding workflows
  - Integrated AI coding assistants into the development workflow to accelerate scaffolding and deployment
  - Engineered modular prompt templates, **guardrails**, adapters, & test oracles for reliability & reproducibility
  - Implemented observability and evaluation dashboards to track latency, errors, token usage, and success rates
- **Project: Production-Scale AI System for Mobile Model Optimization & Deployment**
  - Deployed production ready model & AI platform, scaling to **1M+ daily** predictions for **500K+ users**
  - Introduced **MLOps culture** by setting up CI/CD pipelines with **Amazon SageMaker** tools, reducing integration time for new models from days to hours & increasing deployment frequency from bi-annual to monthly
  - Cut incident response time by 50% through custom monitoring system built with SageMaker **Model Monitor**
  - Achieved **40% faster training** speeds by implementing **distributed training** using **SageMaker DDP**
  - Improved model accuracy by 23% & reduced corrections by 25% through semi-automated continuous retraining
- **Project: AI-Driven Edge Computing System for Multi-Modal Transportation Classification**
  - Designed & developed an **edge-based transportation mode detection system** for iOS platform serving **500K+ users** with **1M+ daily** predictions, achieving **99.3% accuracy** in classifying 15 transportation modes
  - Optimized model footprint by 90% through hierarchical ML architecture while maintaining inference latency
  - Built cross-platform ML pipeline with m2cgen to convert Python models to native Dart, preserving performance
  - Reduced AWS costs by 70% by migrating ML inference to edge devices via **Flutter implementation**
  - Reduced operational costs 60% by implementing local feedback caching to eliminate OpenStreetMap API calls

AI Engineer | Hypothesis AI, IND | Consulting *May 25 – July 25*

- **Project: Multilingual AI Voice Agent for Abandoned Checkout Recovery in E-commerce**
  - Designed scalable **multilingual AI voice agent** using OpenAI & RAG, achieving latency in milliseconds
  - Engineered multilingual (6 languages), context-aware prompts, boosting abandoned checkout recovery by 90%
  - Developed event-driven integration framework with FastAPI, connecting Shopify with third party services and analytics platform (mixpanel), enabling seamless omni-channel customer engagement

AI Engineer (Founding Team) | MindYogi.me, IND | Consulting *Dec 25 – present*

- **Project: Advanced RAG-Enabled Chatbot and Voice Interface for Personalized Wellness Content**

- Developed a scalable serverless architecture for a mental health platform (MindYogi), implementing AI-driven content generation workflows and **text-to-speech synthesis** powered by state-of-the-art models
- Designing and implementing an **advanced agentic RAG chatbot** using LangGraph, incorporating **self-reflective, corrective, and adaptive** retrieval strategies to improve grounding and minimize hallucinations
- Optimizing the retrieval pipeline on Pinecone using recursive, context-enriched hybrid chunking, with reranking, caching, and query rewriting/expansion nodes, with continuous grounding and relevance evaluation via RAGAS
- Optimizing a 10+ node LangGraph pipeline with advanced inference techniques, reducing TTFT to <5 seconds
- Building a low-latency **voice agent** using Fennec ASR, Baseten inference, and Inworld TTS, achieving <2s end-of-speech to first audio frame latency at around $0.50 per hour of voice output
- Deployed real-time chatbot and voice inference as modular REST API services supporting multiple consumers, with low latency, monitoring, and production reliability

Research Software Engineer | Hitachi R&D, Tokyo, JPN | Full Time *Jan 21 – July 23*

- **Project: Distributed Data Management Framework for Microservice Architecture**
  - Developed tool for assisting the implementation of **CQRS** & **Event Sourcing** design patterns
  - Reduced development time by 15% using **Domain Driven Design** software development approach
  - Improved scalability & availability of the developed application using **Kafka** & **EventStoreDB** tools

Patent: *System & method to assist modelling CQRS & ES based application [Submitted]*

Research Assistant | IIT Bombay, Mumbai, IND | Full Time *Sept 20 – Nov 20*

- **Department: Centre for Technology Alternatives for Rural Areas (CTARA)**
  - Developed a **web portal** for farmers to help them in assessing their crop in real-time using remote sensing
  - Estimating and predicting the crop acreage by processing the spatiotemporal satellite images of the crop
  - Performed **Tukey, Fischer, Bonferroni tests** on NFHS-4 data to compare malnutrition in Indian states

PROJECTS

- **AI-Powered Agentic Document Processing System for Automated Form Completion**
  - Developed an agentic, event-driven AI workflow with **LlamaIndex** for automated form filling tasks
  - Integrated **Groq LLM** for dynamic extraction from both structured and unstructured documents
  - Used **Hugging Face** embeddings for vector store indexing and efficient contextual document retrieval
  - Implemented human-in-the-loop feedback system to iteratively refine agent accuracy and user alignment
- **Multimodal AI System with Local LLMs for Document, Image, & Voice Processing**
  - Architected multimodal AI chatbot integrating **Ollama** LLMs and **ChromaDB** for local document processing
  - Implemented semantic document search with **ChromaDB** vector storage for context-aware responses
  - Enabled real-time audio transcription using **Whisper** models and designed intuitive UI with **Streamlit**
  - Developed robust state management with thread-safe **SQLite** pooling, ensuring secure local deployment
- **FarmGPT: MCP-Driven RAG & LLM Agricultural Advisor**
  - Architected modular decision support system using **Model Context Protocol (MCP)** for agriculture
  - Integrated **RAG** servers and **ChromaDB** vector database for semantic search and document retrieval
  - Engineered embeddings & prompt engineering for optimal LLM interaction with **PLLaMa** & agentic LLMs
  - Implemented asynchronous SSE for real-time data streaming and API-first design for actionable farmer guidance
- **Decision Support System for Agriculture Monitoring using Convolutional Neural Network**
  - Developed CNN model for crop classification achieving 94% accuracy using multispectral satellite imagery
  - Created automated data pipeline processing 5TB+ of satellite data for training and prediction
  - Deployed web-based dashboard enabling farmers to monitor crop health and predict yields in real-time

EDUCATION

Indian Institute of Technology Bombay | Technology & Development *Aug 18 - July 20*

- CGPA: 9.3/10 | Machine Learning in Remote Sensing | Advanced Statistics | Satellite Image Processing
- **Publications:** Lonare, A., Maheshwari, B., & Chinnasamy, P. (2022). Village level identification of sugarcane in Sangali, Maharashtra using open source data. Journal of Agrometeorology

Indian Institute of Technology Kanpur | Mechanical Engineering *July 13 - Aug 17*